

Inclusion Techniques for Team Practices in HCI

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ABSTRACT

In tech, women and people of color consistently report being ignored, devalued, and perceived as less competent than men. But if all members of diverse teams do not feel valued and connected, team performance and innovation are undermined. This course introduces practices that interrupt bias and negative interactions while ensuring all are heard during working meetings. We cover effective techniques to structure participation in working team meetings and guide decisions and feedback so all are heard. These techniques arm HCI professionals who are leaders or facilitators, and academics guiding student teams, with ways to ensure inclusion.

CCS CONCEPTS

• **Social and professional topics** → *Computing organizations*.

KEYWORDS

diversity, inclusion, HCI methods, facilitation techniques, organizational culture, bias

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1 DETAILED COURSE DESCRIPTION

1.1 Benefits

Participants will be introduced to effective techniques to ensure that all voices are heard in diverse teams. They will learn about the barriers and biases that women and other underrepresented groups face that undermine their experience of belonging. They will learn how making small changes to how working meetings are run can reduce interpersonal chaos and bias. Armed with examples of how to tune HCI working meetings, they will be better able to design, lead, and facilitate group meetings so that everyone's perspective is considered.

1.2 Intended Audience

This course is relevant to anyone who wants to learn how to ensure that team practices are inclusive and fair, especially HCI professionals who are leaders, team coaches, or facilitators as well as

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academics working with student teams. Anyone tasked with designing practices for inclusivity will benefit.

1.3 Prerequisites

None

1.4 Content

1.4.1 Why this course? In tech, women and people of color consistently report being ignored, devalued, and perceived as less competent than men [Cook 2020; Wynn 2019]. Yet research has definitively shown that a diverse product team improves the quality of products, the breadth of innovation, the success of the organization, and the bottom line [Hunt et al. 2018]. But companies only reap these benefits when all members of the team have a sense of belonging, are engaged in dynamic interactions with other team members, and feel that they are known and valued for their individual contributions. When women feel like they belong, they own the team's mission as their own mission. Then the whole team performs better [Post 2015]. But this is not always or often the case. When men in a team act as an ingroup, women feel pushed away. When women's technical skills are questioned as research shows is usual in tech [Carrigan 2018], women do not feel valued. The situation is even worse for women of color and other intersectional identities; they are even more likely to have their contributions forgotten or misattributed to others [Purdie-Vaughns and Eibach 2008]. As a result, men and women do not communicate and interact in a dynamic way and overall team functioning suffers [Mollema 2005]. Remote work holds some hope that marginalized communities in tech may be valued and empowered [Ford et al. 2019]. But often, when team members are remote, the situation gets worse, not better [Hong et al. 2021; Siegert and Niebuhr 2021; Wang and Roubidoux 2020; Wasche 2021].

The experience of belonging and being valued in the team is one of the most important factors for retaining women and other underrepresented team members in tech [Holtzblatt and Marsden 2018]. The sense of participating in a dynamic team up to something that matters as a valued member is critical to their sense of well-being. This is true for HCI professionals as well. HCI professionals bring their own set of diverse skills to product design. If they do not feel part of the team, team cohesion suffers along with the product. But usability experts feel less valued than those in other tech job roles [Marsden and Haag 2016]. Our research also shows that HCI professionals evaluate the core factors regarding the health of their work experience more negatively than people in other tech job roles. HCI professionals, both men and women, experience less belonging and dynamic participation in teams than other job types [Marsden and Holtzblatt 2018]. And like women in general, HCI professionals are also seen as less technical because they may not have a computer science degree. Diversity is not just an issue of gender, ethnicity,

age, and other identity characteristics. It also includes the diverse job types that make up a product team. Ensuring the cohesion of the team, and that all voices are heard, is important to the success of any HCI professional.

Since team cohesion and the sense of belonging is so important to the psychological health of the team and product success, it is an important intervention point. Improving how people interact and the procedures used in the daily working meetings of diverse teams is a critical starting point for change. Working meetings include, for example, the stand-up and retrospectives meetings in Scrum and design meetings central to HCI professionals' work: brainstorming, design sprints, critiques, and user data review. HCI professionals attend and facilitate many of these meetings. As such, HCI leaders and facilitators are well-positioned to improve or galvanize the team's practices to ensure that all diverse voices are heard on the team – including their own. They can help improve both team cohesion and everyone's experience of belonging by introducing the techniques shared in this course.

The instructors focus on techniques that research and their own work with diverse teams has been shown to encourage equal participation from all members of a diverse team during working meetings. The instructors have been working to develop practical interventions and techniques to improve how diverse teams work for many years. These techniques tune existing practices to interrupt bias that may creep into working sessions. In contrast, bias workshops and other awareness-raising sessions have been tried for many years with little practical result. Unfortunately, women's sense of belonging and feelings of discrimination are largely unaffected by whether their departments complete a diversity training program [Moss-Racusin et al. 2018]. Instead, the best practices taught in this course use procedures that reliably change how people work and interact to interrupt bias [Bohnet 2016; Nordell 2021].

1.4.2 Course description. The course is conducted with a mix of lecture, discussion, and group exercises. This course description is for an in-person session but can be adapted if delivered on-line. We cover the following topics in this order:

- (1) Reflection: Participants talk in pairs about their personal experiences of being heard and participating in brainstorming and critique meetings. We use this exercise as an icebreaker and to raise awareness of the issues we will address.
- (2) Introduction: We start with an overview of the primacy of the team experience for ensuring a positive working environment. We cover the value of diverse teams for creativity, the role of team cohesion, and what contributes to or undermines it. We then cover the attitudes, expectations, and biases that women and other marginalized identities face when they choose a technology profession. We discuss the impact of remote work on these issues. Last, we discuss the power of tuning existing practices to interrupt bias. We talk about why targeting working meetings is a good place for leaders and facilitators to begin.
- (3) Techniques for equal participation in working meetings: We use typical HCI working meetings as the context for discussing participation. They may include brainstorming, design sprints, and critique or review meetings. Techniques

are also relevant for Scrum stand-up and retrospective meetings. We share the participation barriers that diverse team members, early career professionals, or those who are not comfortable speaking in a group face in these group settings. We then discuss key interventions to improve team interaction. These include the importance of allowing people to form and note their ideas independently before group discussion. Next, we introduce the power of round-robin sharing of one idea at a time ensures all are heard. We also emphasize the importance of capturing ideas in a central artifact that all can see and respond to. We use a critique meeting as an example of how to collect useful feedback that can be heard and the way the round-robin format interrupts people who might otherwise grab the floor.

- (4) Structured meetings to mitigate negative interactions. We cover key issues that can arise in working meetings that can undermine the experience and effectiveness of that meeting. Meetings can devolve into chaos, which encourages speaking over each other, accusations, and wasting time on side topics. Chaos also fuels bias behavior. Although people often resist it, our own and other's research [Allen et al. 2015] emphasize the need for clear practices to avoid interpersonal chaos. Well run meetings need for clear goals, agenda, roles, time-boxing, and explicit procedures to ensure that meetings run smoothly. We share the practices of a structured critique meeting and discuss how it clarifies expectations.
- (5) Criteria of Goodness to undermine interpersonal conflict. When working meetings seek out ideas or feedback, they can become personal and communicate devalue. They can also lead the team to commit the "fundamental design error" – generating ideas from their own personal experience. We introduce the need for explicit Criteria of Goodness appropriate to the meeting. These can include key issues in user data, business goals, and design principles. We share Criteria of Goodness for a review meeting and clarify how they are used when providing feedback or comments on the work being reviewed. We also share other examples from different meeting types.
- (6) Application and Conclusion: Participants talk in small groups about how these techniques might be used in their organization in critique and ideation meetings. We collect issues that participants raise and answer questions. We reiterate the need to redesign how we work to change behavior. We reference other ways to tune working meetings and practices.

1.5 Practical work

We use different techniques to get the participants activated and involved. We start with a short reflection in pairs. We encourage participants to think about and take notes on the application of the techniques to their brainstorming and critique sessions during the lecture. This is then used in the exercise after introducing techniques where small groups of participants discuss and apply the techniques to their own organizations. (If the course is on-line we use break-out rooms and Miro boards to facilitate discussion.)

2 INSTRUCTOR BACKGROUND

2.1 Karen Holtzblatt

Karen Holtzblatt is a thought leader, industry speaker, and author. As co-founder and CEO of InContext Design, Karen is the visionary behind Contextual Inquiry and Contextual Design, a user-centered design approach used by universities and companies worldwide. Recognized as a leader in requirements and design, Karen has been twice honored by the ACM SIGCHI. Karen is a member of the CHI Academy and the first recipient of the Lifetime Award for Practice presented in recognition of her impact on the field. Karen is also the Executive Director of WITops, a non-profit dedicated to understanding the issues faced by women in tech and finding practical interventions to retain women and help them thrive. The @Work Experience Framework identifies the six experiences women need in their everyday work experiences. WITops volunteer teams have also developed tested intervention techniques to encourage these experiences available at WITops.org. Karen has over 30 years of experience presenting at conferences, coaching product teams, and advising universities on their HCI training.

2.2 Nicola Marsden

Nicola Marsden is a professor of social informatics at Heilbronn University. She combines insights from psychology, software engineering, design research, and organizational behavior to improve collaboration and foster innovation in technology development. Her research is based on a combination of experience in both academia and industry, often with a gender or cross-cultural perspective. In her transformation work with people, teams, and organizations she offers a balance between scientific knowledge and a practical approach. Throughout her career, Nicola has worked toward bias-free equal opportunity for women. Nicola is vice chair of the National Competence Center Technology-Diversity-Equal Chances in Germany, a non-profit dedicated to ensuring equal opportunity for women and men in STEM education and industry. She has been recognized as the leading expert on gender issues in Scrum in Germany.

3 RESOURCES

Holtzblatt, Karen and Marsden, Nicola. (2022 publication date pending) Retaining Women in Tech: Shifting the Paradigm. Morgan Claypool Publishers.

The introductory chapter can be downloaded at <https://www.morganclaypoolpublishers.com/womenintech/>.

Go to <https://www.witops.org> for all research and interventions related to retaining women in tech. Videos and downloadable resources are available on topics like the @Work Experience Framework of factors that help women thrive; team onboarding and the checklist to guide managers; the critique meeting process; techniques for managing diverse teams; the key behaviors that increase or decrease the experience of valuing and their character posters for awareness, the remote work experience research, and more.

4 ACCESSIBILITY

We will be guided by the suggestions ACM provides and accessibility standards for slides including using alt text for pictures and

avoiding graphics with text to express ideas. We assume that course participants will have a digital version of the slides so that content can readily be seen and enlarged if needed. Exercise material will be in slides.

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